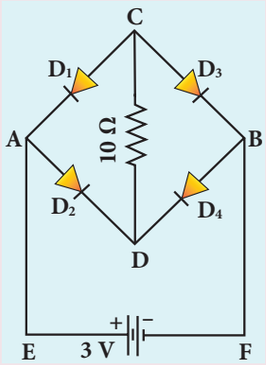
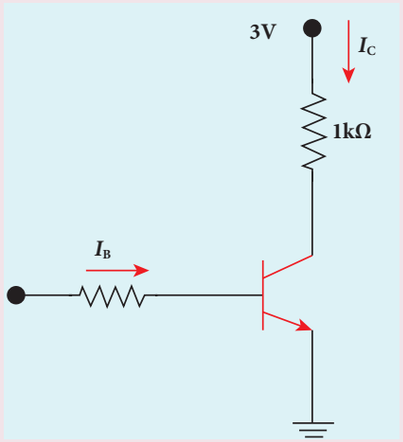


2. Four silicon diodes and a 10 Ω resistor are connected as shown in figure below. Each diode has a resistance of 1Ω. Find the current flows through the 10Ω resistor. [Ans: 0.13 A]

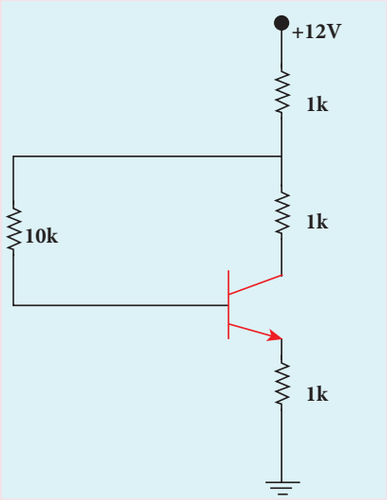


3. Assuming V_{CEsat} = 0.2 V and β = 50, find the minimum base current (I_B) required to drive the transistor given in the figure to saturation. [Ans: 56 μA]

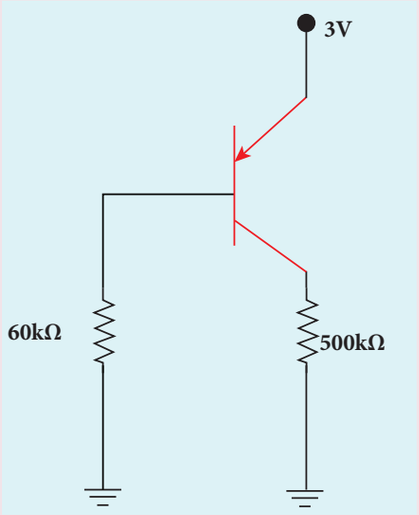


4. A transistor of α = 0.99 and V_{BE} = 0.7 V is connected in the common emitter configuration as shown in the figure. If the transistor is in saturation region, find the value of collector current.

[Ans: 3.73 mA]



5. In the circuit shown in the figure, the BJT has a current gain (β) of 50. For an emitter – base voltage V_{EB} = 600 mV, calculate the emitter – collector voltage V_{EC} (in volts). [Ans: 2 V]



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